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Industrial Automation: From Control to Intelligence

AI is reshaping the automation value pyramid into an hourglass.

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At a Glance

- Industrial automation has begun a structural shift: Value is moving away from control and toward intelligence.
- Profit pools are moving to the top (software, data, AI) and bottom (smart devices) of the stack—leaving the core control technologies in the middle under pressure.
- Legacy advantages are eroding faster than most incumbents expect.
- By 2030, nearly half of industry revenues are expected to rely on AI-enabled offerings.

Industrial automation is no longer about controlling machines—it's about orchestrating intelligence.

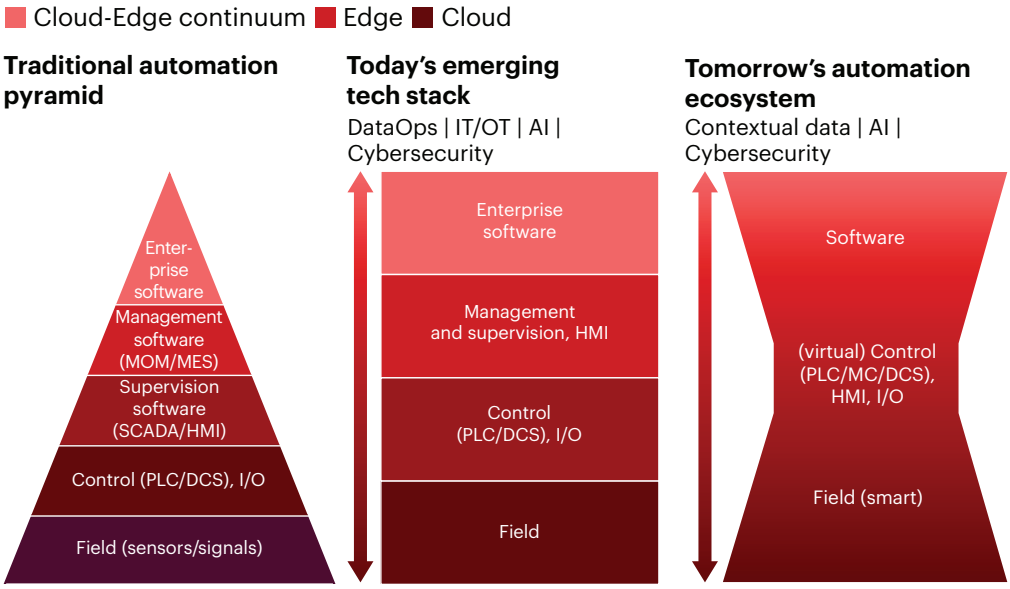
For decades, industry leaders followed a clear logic: Improve production and manufacturing efficiency, quality, and safety through increasingly sophisticated control systems. The economics were obvious. Value sat in proprietary high-performance controllers, tightly integrated systems, and the services and upgrades wrapped around a large installed base.

What's changing is not just automation technology, but where economic value is created in the market.

That logic is now reaching its limits. What's changing is not just automation technology, but where economic value is created in the market. What once looked like a pyramid—value concentrated in control hardware and systems—now looks more like an hourglass, with the middle shrinking and the ends growing *(see Figure 1)*. By the end of the decade, more than 80% of industry profit pools are expected to sit at the two ends of the hourglass, with software and data-driven layers accounting for more than half of the total industry profit pool, and smart field devices capturing an additional 25% to 30% *(see Figure 2)*.

At the top of the stack, value is concentrating in software, data platforms, and AI-enabled workflows. These layers scale faster, carry higher margins, and compound in value as data and use cases accumulate. They increasingly act as the "brain" of industrial operations, translating raw signals into decisions and outcomes. At the bottom, value is reemerging in smart field devices. Sensors, such as machine vision technology, and actuators, such as variable-frequency drives, are no longer passive endpoints. With embedded intelligence, connectivity, and edge computing, they generate data, execute decisions, and continuously improve performance.

Figure 1: As the role of software grows, the tech stack shape is shifting from a pyramid to an hourglass



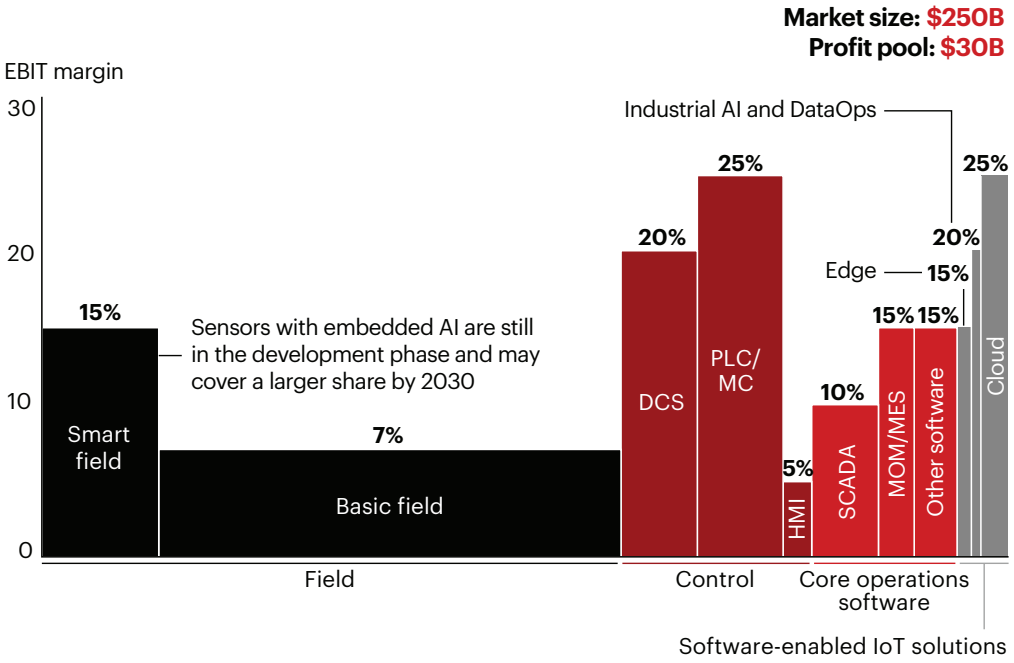
Notes: I/O is input/output module; PLC is programmable logic controller; DCS is distributed control system; SCADA is supervisory control and data acquisition; HMI is human-machine interface; MC is motion control; MOM is manufacturing operations management; MES is manufacturing execution system; IT is information technology; OP is operational technology
Source: Bain analysis

By contrast, the traditional control layer in the middle—programmable logic controllers (PLCs), distributed control systems (DCSs), input/output modules (I/O), supervisory control and data acquisition (SCADA), and their related proprietary software—remains essential but is becoming harder to scale and to differentiate. New entrants are compressing margins by shifting value away from these core controls. By the end of the decade, most industry profit pools will flow to the two ends of this hourglass, away from the center, according to Bain analysis. The implication is stark: Control still matters, but it is no longer the profitable core of the industrial automation industry. The shift is clearly visible today in hybrid verticals such as pharmaceuticals and food and beverage, and will soon be in discrete verticals (e.g., automotive) or process verticals (e.g., chemicals).

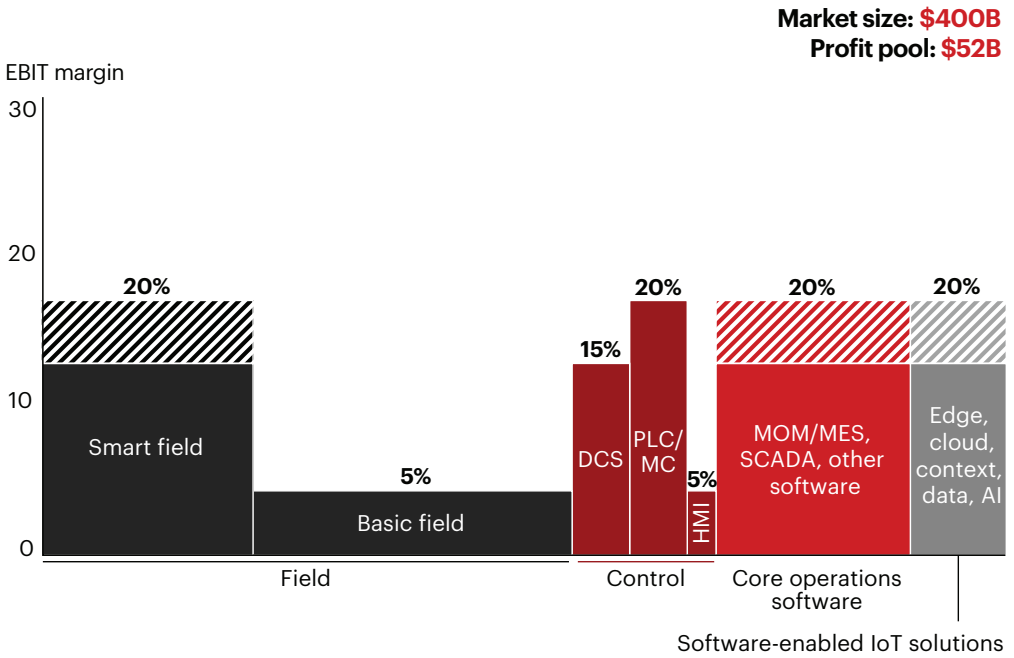
By 2030, nearly half of industry revenues will rely on AI-based solutions, underscoring how value is shifting toward intelligence.

Figure 2: New technologies will significantly shift industry profit pools toward software and digital solutions

2025: Today's emerging tech stack



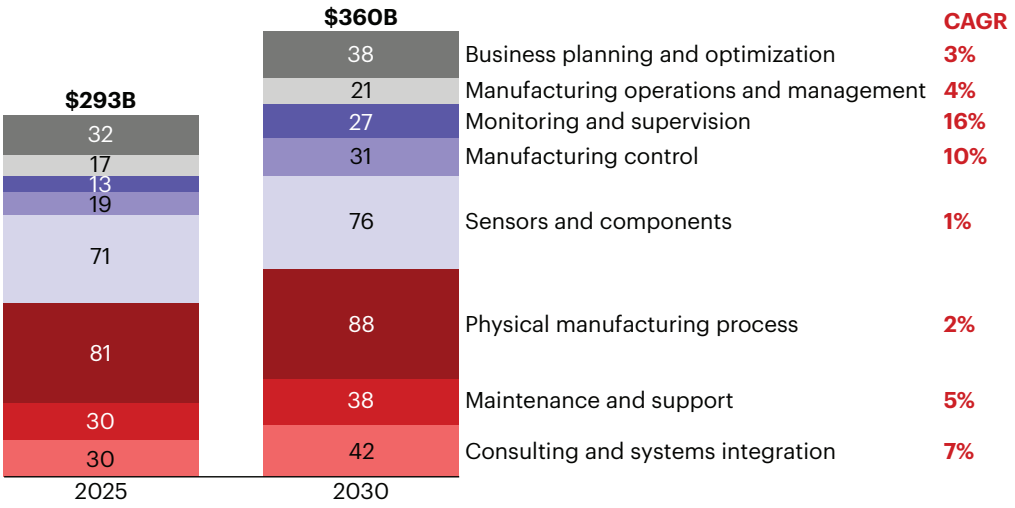
2030: Tomorrow's automation ecosystem



Notes: Market and profit pool size are estimates; field categories include sensors, actuators, drives, industrial robots, conveyors, and machine vision; smart field includes equipment with Internet of Things (IoT)–enabled intelligence, compute, and native connectivity; other software includes historian software, advanced process control (APC), simulation software, and digital twins; I/O is input/output; PLC is programmable logic controller; DCS is distributed control system; SCADA is supervisory control and data acquisition; HMI is human-machine interface; MC is motion control; MOM is manufacturing operations management; MES is manufacturing execution system
Sources: Bain analysis; ARC Advisory Group Automation Report

Figure 3: AI will create new industrial market value of nearly $70 billion over the next five years

Market expansion in billions of dollars, 2025–30



Source: Bain Industrial Automation Executive Survey 2026 (n=26)

Bain research shows that by 2030, nearly half of industry revenues will rely on AI-based solutions, underscoring how value is shifting toward intelligence. AI-enabled solutions alone could unlock up to $70 billion in new market value by 2030, according to Bain's 2026 Industrial Automation Executive Survey *(see Figure 3)*.

Eroding advantage

Most incumbents understand that the industry is going digital. Fewer appreciate how quickly that shift undermines the sources of differentiation they have relied on for decades. Three forces are accelerating the erosion. First, the operating environment has changed fundamentally. Labor shortages are structural: Manufacturing workforces in developed markets are aging rapidly, with more than 40% of US manufacturing employment at firms where at least a quarter of workers are over age 55, limiting the industry's ability to rely on human expertise. Supply chains are being reconfigured for resilience, not just efficiency. Sustainability, cybersecurity, and traceability expectations are rising simultaneously. Legacy automation architectures—optimized for stability and cost—were never designed for this level of volatility.

Second, the sources of differentiation are moving beyond hardware. Control performance is increasingly table stakes. Manufacturers expect systems that can adapt, optimize, and learn over time. In particular, they want production automation technologies that interconnect upstream with design, engineering, and simulation, and downstream with supply chain and distribution systems. As a result, buying decisions are gravitating toward software, data, and use case enablement beyond manufacturing—areas where the installed base of control systems alone offers limited protection.

Third, competition is intensifying from both ends of the stack. Hyperscalers and AI-native players are expanding into industrial software and data platforms. At the same time, aggressive hardware competitors—particularly from China—are compressing margins in controllers and basic automation components, including many categories of sensors and industrial cameras. The result for automation incumbents is pressure from above and below. Switching costs are falling as software decouples from hardware and interoperability improves. Services attached to legacy systems are harder to defend when customers demand continuous improvement rather than periodic upgrades.

The risk for incumbents is not overnight disruption—it's gradual irrelevance. It's a slow drift from most strategic manufacturer partner to component supplier, even while revenues appear stable. This is why the shift feels uncomfortable.

Tomorrow's competitive edge

In the next era of industrial automation, leaders will orchestrate intelligence rather than deploy more technology. It is about how software, data, and smart devices come together—vertically, not horizontally—to solve operational problems. Three patterns stand out as the industry shifts toward software, smart devices, and vertical depth.

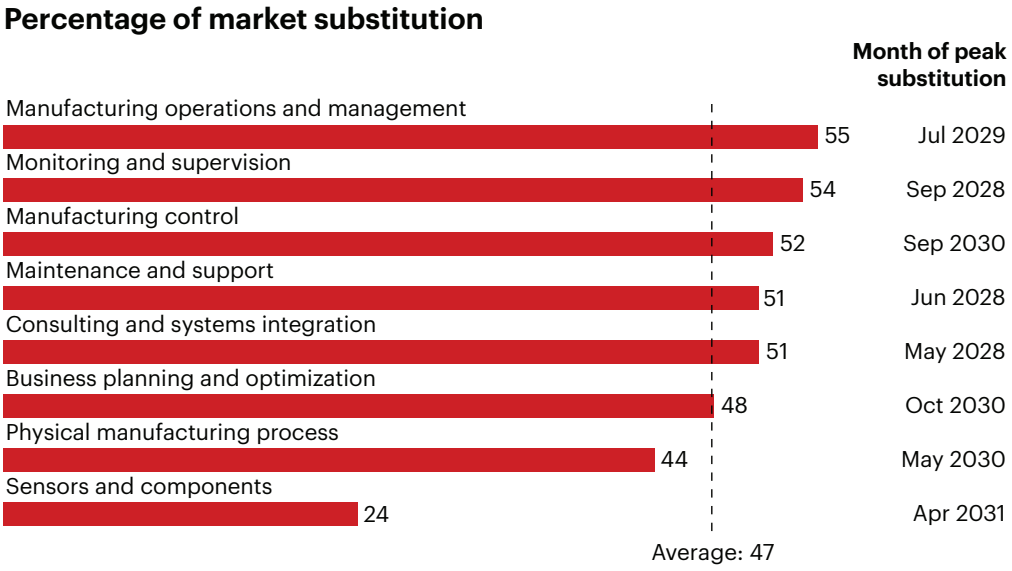
The most important shift is from control logic to decision logic. Traditional automation excels at executing predefined instructions in stable environments. The next wave of value creation comes from systems that continuously decide—prioritizing trade-offs, adapting to variability, and optimizing outcomes across time and assets. AI-native workflows are moving from analytics layers into the operational core, shaping decisions on throughput, quality, energy use, and maintenance in real time. As margins tighten, value accrues to those who own the decision layer—not just the systems that execute instructions. This marks a clear break from the past. Future competitiveness will hinge less on how efficiently processes are automated, and more on how intelligently operations respond when conditions change.

AI's first wave of impact will also be far more concentrated—and time-bound—than many leaders expect. Bain analysis shows that a small number of use cases account for a disproportionate share of AI's upside, led by adaptive robotics, predictive maintenance, and knowledge-based systems.

By 2030, nearly half of industry revenues are expected to rely on AI-enabled offerings, with substitution pressure exceeding 50% in several core use cases *(see Figure 4)*. In these areas, AI is no longer a differentiator—it is a prerequisite for market access. And for the highest-impact use cases, much of that value will materialize within the next one to five years, leaving little room for incremental or experimental approaches.

Software and data are becoming the twin engines of value. Operations platforms, workflow applications, and AI-driven optimization tools are moving from the periphery to the core of industrial systems. They contextualize data, coordinate decisions, and translate complexity into action. Critically, they scale across functions and sites over time, creating economics that hardware alone cannot match.

Figure 4: AI-based solutions are poised to produce half of industrial automation revenues by 2030



Source: Bain Industrial Automation Executive Survey 2026 (n=26)

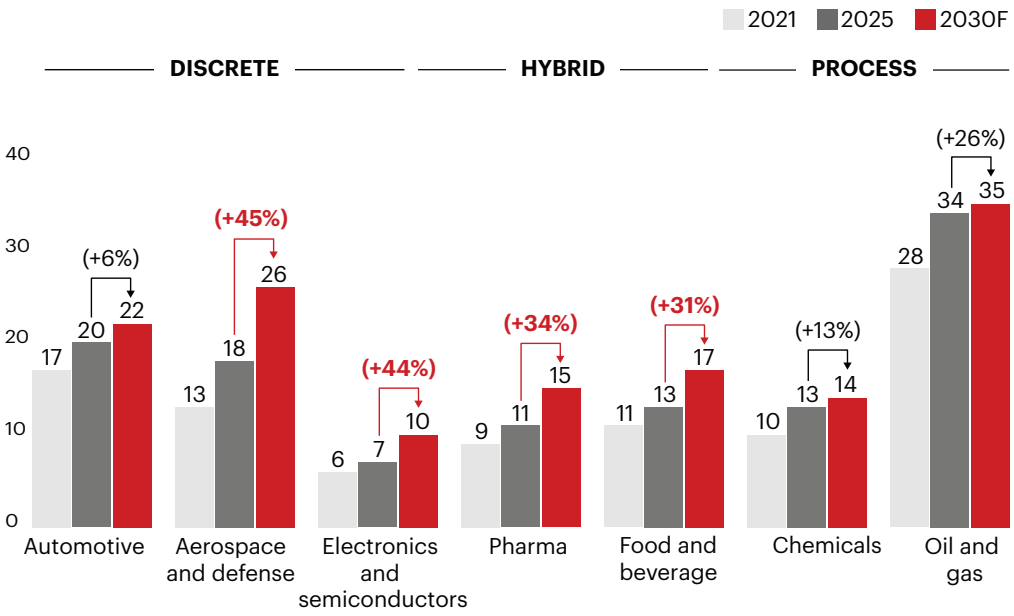
Crucially, the advantage does not come from IT-OT convergence as a technical milestone. Most companies can connect systems, even if the cost of doing so at scale is often prohibitive, especially when measured against each individual use case. Far fewer translate integrated data into faster, better operational decisions. What sets leaders apart is operational convergence—data, governance, and workflows designed to cut across production, quality, maintenance, planning, and energy, and increasingly link to design (i.e., product lifecycle management systems) and to distribution (i.e., supply chain management systems). When insights flow directly into execution, organizations move from reporting performance to shaping it. For leadership teams, this is not only an architecture debate but also an operating model challenge.

Smart devices are part of the decision stream. Intelligence is moving closer to the physical process. Machines and sensors increasingly preprocess data, make local decisions, and collaborate with higher-level systems. This reduces latency, improves resilience, and unlocks new use cases—from predictive quality to autonomous maintenance.

Vertical depth is a new source of differentiation. Industry-specific solutions that embed process knowledge, data semantics, and regulatory requirements will fuel future growth. Nearly 60% of incremental industry growth toward 2030 is expected to come from vertical-specific offerings rather than horizontal platforms *(see Figure 5)*. Food and beverage players care about traceability and hygiene. Battery and automotive manufacturers care about yield, throughput, and rapid reconfiguration. Life sciences demand validation and compliance as core features, not add-ons.

Figure 5: Hybrid manufacturing industries are expected to allocate a higher share of spending to automation

Industrial automation spending in US dollars, per $1,000 of value added in each industry sector



Notes: Discrete industries produce individual, countable units (e.g., bottles, furniture, automobiles), typically through assembly-based manufacturing processes; process industries operate through continuous production flows (e.g., paper, oil, chemicals), where output is not easily separated into distinct units; hybrid industries combine both models—beginning with bulk or continuous processing and followed by discrete assembly or finishing steps
Sources: Market participant interviews; Industrial Automation Market Forecast to 2030 (Market Research Future); Global Industrial Automation Market 2025–2029 (Technavio); Top 15 Growth Opportunities in Industrial Automation 2024 (Frost & Sullivan); S&P Global

As a result, growth and value creation are concentrating in verticalized stacks that combine software, data, and devices into integrated solutions. Competitive advantage increasingly depends on understanding how an industry actually runs—not just how its machines are controlled.

This is also where business models are shifting. Recurring revenues, outcome-based contracts, and lifecycle value are becoming more important than one-time sales. Providers that can measure performance, share risk, and stay embedded in operations capture disproportionate value.

As intelligence becomes continuous, value creation is likely to shift from point solutions to lifecycle orchestration. Customers increasingly reward partners who stay engaged beyond commissioning—improving performance in ramp-up, operations, and optimization. This favors platforms and recurring engagement models, but more importantly, it favors firms that can measure impact and remain accountable for outcomes. Those who "own the system" over its lifecycle—shaping how it learns and evolves—deliver value year after year. For leadership teams, the growth objective shifts from winning more projects to securing deeper, long-running roles in customer operations.

The common thread across all three trends is orchestration. Tomorrow's leaders will be companies that can connect layers, align incentives, and continuously improve outcomes. No company can win alone. As automation evolves toward autonomy, value creation increasingly spans ecosystems of hyperscalers, automation vendors, software specialists, and integrators. The strategic question is not whether to partner, but how to orchestrate.

In the next phase of industrial automation, the winners will coordinate intelligence across partners as effectively as they do across machines. Early leaders are already seeing results. In our experience, companies that orchestrate data, software, and smart devices at scale are achieving productivity gains of 30% to 50%, maintenance cost reductions of up to 35%, and longer asset lifetimes.

Leadership agenda

This shift cannot be addressed with incremental upgrades or isolated digital initiatives. It requires clear strategic choices.

Choose where to win. Emerging automation leaders are prioritizing the industry verticals and choosing the layers of the hourglass where they intend to win. They are also staking out what business areas to defend, where to partner, and where to step back. Trying to protect the entire stack spreads investment too thin.

Go deep in chosen verticals. Process expertise, industry-specific data models, and tailored use cases are becoming decisive. Horizontal scale alone is no longer enough.

Treat software and data as core assets, not extensions. This means using modern architectures, creating an AI-enabling data strategy, and embedding AI in workflows—not layering pilots on top of legacy systems.

Redesign the commercial model to focus on lifetime value. Successful companies will redesign metrics, incentives, and go-to-market approaches to focus on outcomes rather than transactions. This often creates short-term friction—but in the long term, it will increase returns.

Reinvent the ecosystem. No company can win alone. Future leaders will partner with hyperscalers, integrators, and specialists, ensuring that roles are clear and that the value captured is deliberate. The companies that move early will shape the rules of the next era.

Industrial automation is entering a new phase. Factories are becoming adaptive systems that sense, learn, and act across the value chain. Productivity gains of 30% to 50%, lower maintenance costs, and longer asset lifetimes are within reach—but only for those that coordinate intelligence across machines.

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